AI-POWERED STORE INTELLIGENCE

CCTV Retail Analytics & POS Receipt Correlation Engine

Store Location:	Brigade Road, Bangalore (ST1008)
Team Name:	Purplle Hackathon Team 2026
Key Features:	YOLOv8s + ByteTrack + OSNet ReID + 6-State FSM + POSCorrelator V2
Status:	Production-Hardened, Fully Validated, 82 Tests Passing

PHYSICAL RETAIL'S "BLACK BOX"

The massive data gap between physical store operations and e-commerce

- **E-Commerce Tracking vs. Physical Blindspot**
  Online stores track every click, scroll, cart addition, and checkout drop-off. Physical stores only know total footfall and final sales—missing the visitor journey in between.

- **Missing Shelf-Level Insights**
  No visibility into brand engagement, display zone dwell times, and how zone visits translate into billing conversions.

- **Undetected Store Bottlenecks**
  Checkout queue line spikes and dead display zones (displays getting zero attention) remain invisible in daily aggregates.

- **Attribution Failure**
  Inability to correlate specific visitor pathways with actual POS basket transactions to calculate brand-level conversion rates.

THE CCTV STORE INTELLIGENCE PLATFORM

Leveraging existing hardware to unlock web-style retail analytics

- **Zero Hardware Upgrades**
  Runs directly on existing CCTV security camera streams, using standard CPU processing for edge deployment.
- **Resource-Driven Design**
  Ingests the physical store layout blueprint and real POS transaction logs as the single source of truth for conversions.
- **Privacy-First Architecture**
  Computes coordinates and tracks at the edge. No PII (Personally Identifiable Information) or facial images are stored, emitting only anonymous events.
- **Unified Dashboard & Real-Time Alerts**
  Presents Plotly funnel conversions, brand shelf heatmaps, queue dwell times, and proactive anomaly warnings on a live dashboard.

TECHNICAL PIPELINE & DATA FLOW

The 6-stage edge processing sequence

1. YOLOv8s Detector

Locates person bounding

boxes on CPU (30+ FPS)

2. ByteTrack Tracker

Kalman filter state

recovery for occlusions

3. OSNet ReID Gallery

Lightweight appearance

handover (similarity >0.75)

4. 6-State FSM

Tracks Entry -> Zone ->

Queue -> Billing -> Exit

5. POS Correlator V2

Matches invoices to exits

via brand overlap & time

6. FastAPI & Streamlit

Live REST endpoints &

dark-mode visualization

STORE LAYOUT & SEMANTIC ZONES

Mapping camera pixels to physical retail brands

- **Store Layout Parsing**
  Loads `store_config.json` containing normalized polygon coordinates of 20 named display zones (Lakme, Minimalist, Aqualogica, Loreal, etc.) and entry/exit lines.

- **Dynamic Coordinate Scaling**
  Automatically scales normalized coordinate polygons to the actual resolution of each camera frame, ensuring pixel-perfect polygon checking.

- **Shapely Geometry Checks**
  Utilizes Shapely's `within` and `contains` functions to track when a customer's bounding box centroid crosses into a brand zone.

- **Zone-Level Dwell Time**
  Tracks individual visitor time within each display zone to filter out brief walk-throughs from active brand engagement events.

HYBRID STAFF DETECTION ENGINE

Filtering employees to protect business metric integrity

- **The Necessity of Filtering**
  Employees move constantly across shelves and dwell for hours. Including them in statistics inflates footfall and destroys conversion rates.

- **Pre-Opening Arrival Check (+2 points)**
  Identifies tracks first observed before `STORE_OPEN_HOUR` (e.g. 09:30 AM startup).

- **Session Dwell & Frame Density (+2 points)**
  Flags tracks present in the store for over 4 hours or visible in >30% of total processed frames.

- **Uniform Color Matching (+2 points)**
  Computes dominant HSV color within the target bounding box to check against the blue-ish staff uniform range (Hue 100-140).

- **Optimal Calibration Results**
  Achieved **100% precision and 100% recall** in validating 12 labeled paths (3/3 staff correctly filtered, 0 customers misclassified).

POS CORRELATION ENGINE V2

Connecting transaction receipts to shopper journeys

- **Temporal Proximity Window**
  First isolates POS transactions matching the store ID and occurring within a `±5 minute` window of each visitor's exit timestamp.

- **Brand Overlap Matching**
  For multiple candidates within the window, the engine intersects the list of zones the visitor engaged with against the brands purchased on the invoice.

- **Multi-Exit Conflict Resolution**
  If brand overlaps are equal, the transaction is attributed to the session with the closest exit timestamp (temporal tie-breaker).

- **Grounded Financial Reporting**
  Ensures all metrics, funnel conversion rates, and revenue mappings are computed dynamically from actual transaction files—no estimates.

OPERATIONAL ANOMALY ENGINE

Proactive detection of store inefficiencies

- **Checkout Queue Spike Alert**
  Triggers when cash counter queue depth exceeds a moving average baseline by more than 2.0 standard deviations (2.0σ), warning of bottlenecks.

- **Display Dead Zone Alert**
  Flags display areas (e.g., Alps Goodness) that receive zero visitor engagement events for a configurable duration (e.g. 60+ minutes).

- **Conversion Drop Alert**
  Detects when the hourly conversion rate drops below a preset performance target (e.g., a 15% drop compared to typical store baseline).

- **Immediate Operational Value**
  Allows store managers to dynamically reallocate floor staff to checkout lanes or adjust product placements on dead shelves.

VALIDATION MODE 1: REAL CCTV RUN

Validating the computer vision and tracking stack on raw footage

Videos Processed	Total Raw Events	Unique Visitors
5 Cameras (Brigade Road)	**326 Events Ingested**	**131 Customers**

Zone Engagements	Top Performing Zone	POS Correlation Matches
54 Visitors (41.2%)	**FOH (Front of House)**	**0 Matches (Expected)**

Validation Mode 1 proves that the YOLO + ByteTrack + OSNet ReID pipeline works correctly on real, uncontrolled video streams. Zero POS matches are expected because the video timestamps (May 31st) and POS transaction database (April 10th) are deliberately misaligned to represent a real-world testing partition.

VALIDATION MODE 2: BUSINESS FUNNEL DEMO

Validating POS correlation and funnel analytics

Simulated Customers

40 Visitors

POS Transactions

24 Unique Invoices

Attributed Revenue

Rs. 31,269.76

POS Match Rate

87.5% (21/24 Invoices)

Funnel Conversion

52.5% (21/40 Sessions)

Queue Dwell Drip

7.7% Checkout Abandonment

Validation Mode 2 aligns the database event timestamps with the POS transaction logs. This proves that the end-to-end analytics engine—FSM session tracking, queue occupancy, checkout abandonment, and POS attribution—functions perfectly under temporal synchronization.

PERFORMANCE BENCHMARKS & COVERAGE

Proof of a production-ready, well-defended codebase

- ### Execution Speed
  Achieves **3.8 FPS** average processing speed on a standard CPU (YOLO detection, ByteTrack tracking, and OSNet ReID embedding extraction combined).

- ### Memory Footprint
  Maintains a peak RAM consumption of only **476.7 MB** during active execution, making it viable for low-cost edge hardware.

- ### API Latency
  Average event ingestion request latency is **16.3 ms** on the FastAPI backend, easily supporting concurrent real-time inputs.

- ### 100% Test Success
  82/82 tests pass successfully across the test suite, verifying all edge cases, state transitions, and database schema mappings.

- ### Coverage Gate Met
  Achieved **80.71% test coverage** across the entire source codebase, exceeding the strict 80% hackathon gate.

PRODUCTION ROADMAP & SCALE

The path to deploying across thousands of stores

- **Appearance Gallery Upgrades**
  Integrate HS/HSV color histograms to supplement OSNet embeddings, boosting ReID precision under extreme lighting changes.

- **Cross-Camera Spatial Calibration**
  Incorporate overlapping field-of-view topology constraints to restrict ReID candidate searches to adjacent camera nodes, reducing ID switches.

- **NTP Clock Synchronization**
  Implement hardware-level Network Time Protocol (NTP) syncing on security NVRs to narrow the POS correlation window to ±60 seconds.

- **Query Performance Indexing**
  Transition from SQLite to PostgreSQL + TimescaleDB for hyper-efficient time-series queries and interval-tree POS correlation index matching.